

Additional Problems

Problem Set

Machine Learning Advanced Track

Question 1: The Optimality of AdaBoost's Weight Update

AdaBoost minimizes the exponential loss function

$$L(y, f(x)) = \exp(-yf(x))$$

where $y \in \{-1, 1\}$ and $f(x)$ is the additive ensemble model. At iteration t , we want to add a weak learner $h_t(x) \in \{-1, 1\}$ with a scalar weight $\alpha_t > 0$ such that the ensemble becomes $F_t(x) = F_{t-1}(x) + \alpha_t h_t(x)$.

Given the observation weights $w_i^{(t)} = \exp(-y_i F_{t-1}(x_i))$, prove that the optimal step size α_t that minimizes the exponential loss on the training data is exactly:

$$\alpha_t = \frac{1}{2} \ln \left(\frac{1 - \epsilon_t}{\epsilon_t} \right)$$

where $\epsilon_t = \frac{\sum_{i: h_t(x_i) \neq y_i} w_i^{(t)}}{\sum_{i=1}^N w_i^{(t)}}$ is the weighted error rate of the weak learner.

Question 2: Gradient Boosting with Cubic Loss

Standard Gradient Boosting Machines (GBM) for regression typically use the Mean Squared Error (MSE) loss, $L(y, F) = \frac{1}{2}(y - F)^2$. Suppose we are dealing with a dataset where we want to disproportionately penalize larger errors, but we want a differentiable function that is steeper than a quadratic. We define a new custom "Cubic Absolute Loss" function:

$$L(y, F) = \frac{1}{3}|y - F|^3$$

- (a) Derive the expression for the pseudo-residuals r_{im} at iteration m for the i -th data point using this custom loss function.
- (b) Let R_{jm} be a terminal node (leaf) in the regression tree built at iteration m . Write down the exact mathematical optimization problem needed to find the optimal terminal node value γ_{jm} .
- (c) **Numerical Step:** Suppose at iteration $m = 1$, the current model predicts a constant $F_0(x) = 5$ for all data points. We have three training points with true targets $y_1 = 2$, $y_2 = 5$, and $y_3 = 8$. Calculate the exact pseudo-residuals $r_{1,1}$, $r_{2,1}$, and $r_{3,1}$ that the first regression tree will attempt to fit.

Question 3: Two Iterations of AdaBoost

Consider a binary classification problem with a tiny dataset of 5 points. The feature X is a 1-dimensional continuous variable, and the target $Y \in \{-1, 1\}$.

We use decision stumps of the form $h(x) = \text{sign}(t - x)$ (meaning $h(x) = +1$ if $x < t$, and -1 if $x > t$) as our weak learners.

Observation (i)	1	2	3	4	5
Feature (x_i)	1.0	2.0	3.0	4.0	5.0
Target (y_i)	+1	+1	-1	+1	-1

- (a) **Iteration 1:** Assume uniform initial weights $w_i^{(1)} = 0.2$. Find the optimal threshold $t_1 \in \{1.5, 2.5, 3.5, 4.5\}$ that minimizes the weighted training error. Calculate the error ϵ_1 , the stump weight α_1 , and compute the unnormalized updated weights for all 5 points.
- (b) **Iteration 2:** Normalize the weights from Iteration 1 to get $w_i^{(2)}$. Using these new weights, determine the weighted error if we were to select the threshold $t = 3.5$.

Question 4: Outliers in Boosting

Explain mathematically and conceptually how Extreme Outliers affect:

- (i) AdaBoost (using exponential loss).
- (ii) Gradient Boosting for Regression (using standard MSE).
- (iii) Gradient Boosting for Regression (using Mean Absolute Error, MAE).

Which of these three configurations is the most robust to a heavily corrupted target variable, and why? Refer to the gradient of the loss function in your answer.